solely upon the initial and end states of the atomic nucleus. Generally speaking, all nuclei of the same kind have the same energy, but in the case of beta rays the situation is different from that of the alpha rays. Any homogeneous radioactive substance will emit beta particles with all possible velocities, from an upper limiting velocity down to quite low speeds. It seems that the energy corresponding to this upper limit is identical with the difference between the energies of the atom in its initial and end states. The occurrence of slower particles would, however, contradict the energy principle, unless the energy missing from the individual beta particles were removed from the nucleus in some other way. This brings us to the theory of Pauli, that with every beta particle another particle leaves the nucleus, carrying the difference in energy. The sum total of the energy is always constant, and this total energy is shared by the beta particle and this new particle, in accordance with definite statistical laws.

This new particle must be devoid of all electric charge, for otherwise it would be impossible to explain the fact—confirmed experimentally in every instance—that when a beta radiation takes place, the nuclear charge increases by one unit. The absence of an electric charge on this new particle is indicated also by the fact that these particles cannot be observed in the cloud chamber. Since this new particle is electrically neutral, and its mass is certainly very small, it is called the *neutrino.* According to the evidence of every experiment hitherto performed, the mass of the neutrino is smaller than the mass of an electron. But whether it is actually 0, like the rest mass of the photon, cannot be stated with any certainty as yet.

II. ARTIFICIAL NUCLEAR TRANSMUTATIONS

We have thus described those processes in which the atoms themselves provide us with clues to the properties of their nuclei; now we shall proceed immediately to consider those experiments aimed at gaining more exact and detailed information concerning the nucleus, by means of interference from outside. Again, it was Rutherford who took the first step in this direction, too. He discovered the proper tool for the artificial transmutation of one atom into another—bombardment of atoms by alpha particles. In 1919, he achieved by this method the transmutation of an element: he turned nitrogen into oxygen. But let it not be assumed that this method offered the means of transmuting ponderable quantities. The transmutation affected a very small number of atoms only; but, of course, this does not detract from the fundamental significance of this discovery.

Rutherford found that a certain type of radiation, consisting of positively charged hydrogen atoms—nuclei of hydrogen—was emitted when nitrogen